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From Network Fundamentals to Macro-Financial Integration: The Evolving Predictability of Bitcoin Returns

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Abstract

We examine how the sources of return predictability in Bitcoin evolve as the market integrates with traditional finance. Using daily data from 2014 to 2025, we identify regimes with the sparse change-point method of Dufays and Rombouts [2020]. In each regime, we assess horizon-specific predictability using Breitung–Candelon frequency-domain Granger causality and transfer entropy. Predictors include macro-finance variables (equities, volatility, the dollar, commodities, and Treasury yields and term structure) and crypto-native measures constructed from blockchain and derivatives data. Crypto-native valuation and positioning variables, in particular the market-to-realized value ratio, funding rates, and open interest, predict Bitcoin returns in every regime. Macro-finance variables gain forecasting power after 2019 and become most informative during the 2022–2023 tightening episode, when nonlinear information flows from traditional markets to Bitcoin are strongest. The results imply that Bitcoin’s return dynamics are shaped by internal valuation and leverage cycles throughout, while its exposure to global risk conditions is episodic and concentrated in stress periods.

Keywords: Bitcoin, cryptocurrencies, on-chain metrics, frequency-domain Granger causality, transfer entropy, structural breaks, systemic risk

JEL classification: G12, G15, C22, E44

1. Introduction

Bitcoin, the first decentralized cryptocurrency to be adopted on a large scale, remains the largest digital asset by market capitalization, accounting for roughly half of the total cryptocurrency market as of late 2024¹. Since its inception in 2009, Bitcoin has had substantial spillover effects on other cryptocurrencies [Corbet et al., 2019], establishing itself as the bellwether of the crypto market. However, Bitcoin’s fundamental

¹Bitcoin dominance measures Bitcoin’s share of the total cryptocurrency market capitalization. Earlier digital money experiments (DigiCash, e-gold) relied on centralized entities, making them vulnerable to regulatory shutdown. Bitcoin’s innovation was solving this problem through decentralization.

characteristics and market dynamics have undergone profound changes over its 15-year history.

Initially conceived by Nakamoto [2008] as a peer-to-peer electronic cash system—a decentralized alternative to fiat currencies operating independently of central authorities—Bitcoin’s actual market behavior has departed from this vision. In early empirical work, Bitcoin is characterized primarily as a speculative asset exhibiting bubble-like behavior [Baur et al., 2018]. Subsequent research explored its potential as a portfolio diversifier and a hedge against traditional assets [Dyhrberg, 2016, Bouri et al., 2017], its safe-haven properties during market stress [Conlon and McGee, 2020], and its viability as an alternative to gold [Baur et al., 2018, Klein et al., 2018].

The cryptocurrency ecosystem has expanded from a fringe experiment into a multi-trillion-dollar market. Bitcoin’s market capitalization grew from a few million dollars in 2010 to approximately USD 2 trillion in late 2024, when its price surpassed USD 100,000. This growth has been reinforced by institutional adoption through spot Bitcoin exchange-traded funds (ETFs)—approved by the United States Securities and Exchange Commission in January 2024—and by integration into corporate treasuries, with firms like MicroStrategy holding over BTC 200,000. While the entry of thousands of alternative cryptocurrencies has produced a diverse digital-asset landscape, Bitcoin continues to anchor the system, serving as an entry point for traditional capital and as the industry benchmark. It is increasingly characterized as “digital gold,” valued for its scarcity, neutrality, and resilience.

This paper is motivated by a central question: as Bitcoin has matured, have the primary drivers of its returns shifted from network-based activity to macro-financial determinants? In its early years, Bitcoin was viewed primarily as an experimental token with speculative appeal but limited connection to conventional assets. Early studies documented weak correlations with financial variables, reinforcing the notion of Bitcoin as a segmented market [Baur et al., 2018]. More recent evidence suggests a change in this relationship, particularly following the COVID-19 pandemic and subsequent monetary interventions [Corbet et al., 2020, Caferra and Vidal-Tomás, 2021]. The approval of spot Bitcoin ETFs, increased hedge-fund participation, and the emergence of regulated vehicles further highlight Bitcoin’s growing entanglement in the global financial cycle.

Building on this motivation, we examine how the predictive content of macro-financial factors—equity indices, volatility measures, commodity prices, and interest-rate variables—compares to that of on-chain metrics such as network usage, exchange flows, leverage, and funding dynamics. By analyzing distinct market regimes, we assess whether Bitcoin now resembles “digital gold” with the characteristics of a potential safe-haven or behaves as a high-beta, liquidity-sensitive risk asset. Our approach situates Bitcoin within the broader evolution of the crypto-market structure (derivatives

adoption, venue consolidation, ETFs) and the increasing integration of digital assets into assessments of global liquidity conditions.

Our contribution is threefold. First, we undertake a regime-by-regime comparison of linear (frequency-domain Granger causality) and nonlinear (transfer entropy; TE) predictability, documenting how the information structure of Bitcoin returns has evolved across five distinct market phases from 2014 to 2024. To segment the regimes, we apply the structural break approach developed by Dufays and Rombouts [2020]. Second, we demonstrate that on-chain valuation and leverage metrics—particularly the market-to-realized value (MVRV) ratio, funding rates, and open interest—exhibit persistent predictive content for Bitcoin returns across all regimes, operating alongside, rather than displacing, macro-financial factors. Third, we show that the nonlinear coupling of Bitcoin and traditional financial variables intensifies during periods of stress (notably 2022–2023), while remaining limited in lower-volatility environments—a pattern consistent with regime-dependent price dynamics.

Despite a growing literature on cryptocurrency markets, few studies combine macro-financial variables and crypto-native on-chain metrics within a unified frequency-domain and information-theoretic causality framework across multiple regimes. Financial economists deploy sophisticated tools to assess causal inference, but often overlook blockchain-native data; computer scientists analyze on-chain data but typically avoid econometric causal-inference frameworks. Our analysis bridges this divide by demonstrating that on-chain fundamentals contain predictive information about market outcomes that operate through linear and nonlinear channels, with implications for asset pricing, risk management, and regulatory monitoring. By documenting that nonlinear methods capture state-dependent predictability missed by linear models, we show that Bitcoin’s unique characteristics—radical transparency combined with episodic volatility—benefit from econometric approaches that synthesize insights from distributed systems and financial econometrics.

The remainder of the paper is organized as follows. Section 2 provides background information on Bitcoin’s structural evolution, a bibliometric review, and a review of the relevant literature. Section 3 describes the frequency-domain Granger causality and TE methodologies, and the data description. Section 5 presents the empirical results of our comparison of linear and nonlinear predictability across market regimes. Section 6 concludes.

2. Background

2.1. Structural Break Analysis

Since its introduction in 2008 through the Bitcoin white paper, the cryptocurrency has undergone a substantial evolution. We identify and characterize the key structural changes in its dynamics over time by conducting a structural break analysis following the approach proposed by Dufays and Rombouts [2020]. This allows us to detect shifts in Bitcoin’s underlying behavior and investigate the economic and market forces driving these transitions.

We employ the sparse change-point (CP) framework developed by Dufays and Rombouts [2020] to identify structural breaks in the time series while avoiding over-parameterization. Unlike traditional structural break models that assume all parameters shift simultaneously at detected break points, the sparse CP model allows heterogeneous parameter changes, with only a subset of parameters changing at each break point.

Consider an $\text{AR}(p)$ process with K breaks at unknown dates $\tau = (\tau_1, \dots, \tau_K)$:

$$y_t = \alpha_{k,0} + \sum_{j=1}^p \alpha_{k,j} y_{t-j} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_k^2), \quad t \in [\tau_{k-1} + 1, \tau_k] \quad (1)$$

Following Dufays and Rombouts [2020], we adopt a cumulative sum parameterization where regime k parameters evolve as $\alpha_{k,j} = \alpha_{1,j} + \sum_{\ell=2}^k \Delta\alpha_{\ell,j}$ and $\sigma_k^2 = \sigma_1^2 \prod_{\ell=2}^k \lambda_\ell$. This allows individual parameters to remain constant ($\Delta\alpha_{\ell,j} = 0$ or $\lambda_\ell = 1$) while others change, avoiding the restrictive assumption that all parameters shift simultaneously at each break.

We specify mixture-of-two-uniforms (2MU) priors on parameter changes $\Delta\alpha_{\ell,j}$ and variance multipliers λ_ℓ to induce sparsity:

$$2\text{MU}(x|\delta) = \frac{1}{2\delta} \left[\mathbb{I}_{[1-\delta, 1+\delta]}(x) + \mathbb{I}_{[-\delta, \delta]}(x) \right] \quad (2)$$

This prior shrinks negligible changes toward zero while allowing substantive changes to be estimated without excessive bias. Standard normal and inverse-gamma priors are used for initial regime parameters $\alpha_{1,j}$ and σ_1^2 .

Given the multi-modal posterior induced by discrete break points, we implement a Sequential Monte Carlo (SMC) sampler with tempering [Del Moral et al., 2006]. The algorithm uses $N = 500$ particles, a geometric cooling schedule $\phi_m = (m/20)^3$, and block-wise Metropolis-Hastings mutations. We initialize break locations using Ng et al.

[2021]’s information criterion and work with standardized data for numerical stability.

We estimate the posterior probability that parameter α_j changes at break τ_k as $\Pr(|\Delta\alpha_{k,j}| > 0.1|y)$, considering changes substantive if this probability exceeds 0.5. Parameter estimates use posterior means with 95% credible intervals from the weighted particle distribution.

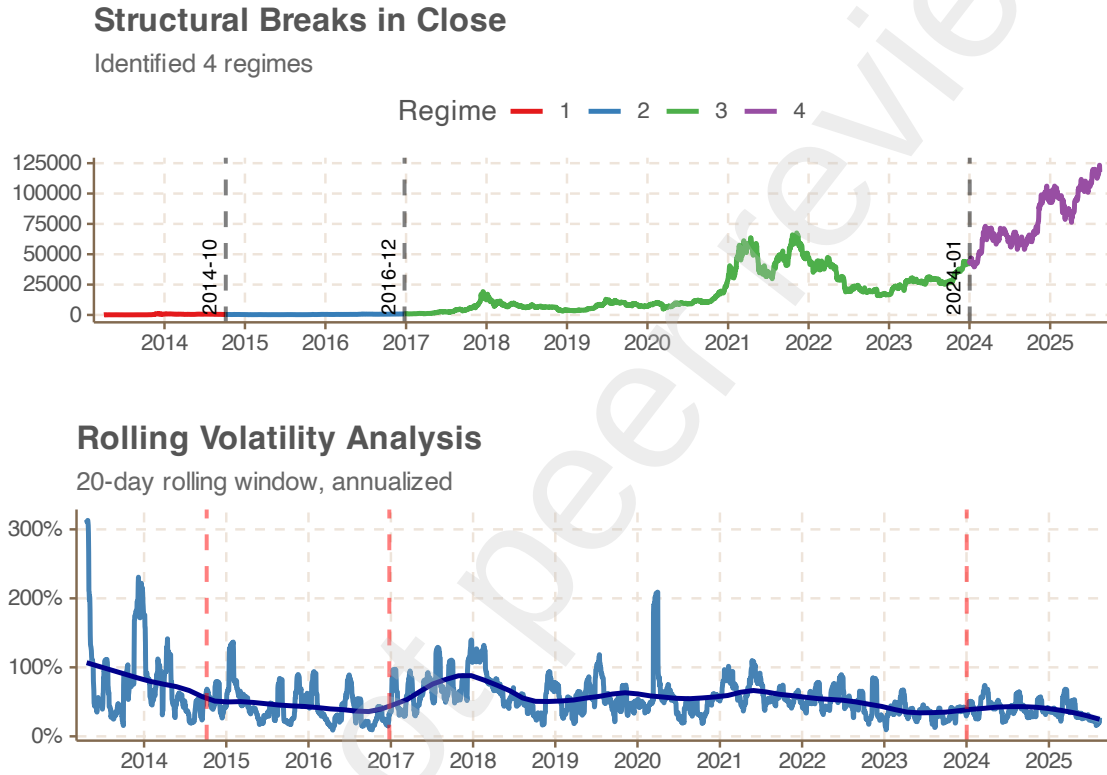


Figure 1

2.2. Structural Break Regimes

Our structural break analysis identifies four distinct regimes in Bitcoin’s price history. The first regime, spanning from Bitcoin’s inception in 2010 until late 2014, corresponds with the initial “experimental phase.” During this period, Bitcoin was primarily used in niche online communities and was characterized by remarkably low liquidity and limited exchange infrastructure. The 2014 collapse of Mt. Gox, then the dominant exchange, handling over 70% of global trading volume, marked the culmination of this

regime². The exchange's bankruptcy, which was triggered by a massive hack, severely undermined investor confidence, prompting a sharp price correction and heightened regulatory scrutiny. This episode effectively ended Bitcoin's formative stage and its fragile and highly concentrated market.

The second regime, spanning from late 2014 to December 2016, can be described as the early phase of institutionalization and market consolidation. Following the Mt. Gox collapse, the ecosystem migrated from fragile, concentrated venues to more robust exchanges such as Coinbase and Bitstamp. This move attracted capital and institutional investors, helping restore confidence in the asset. This process was reinforced by various regulatory events: in 2015, the New York Department of Financial Services introduced the BitLicense, the first crypto-specific regulatory framework in the United States, while the launch of Gemini (by the Winklevoss twins) and the Grayscale Bitcoin Trust (GBTC) created regulated gateways for institutional investors. At the technological frontier, with the launch of Ethereum in July 2015, investor attention was extended to blockchain-based assets, and the crypto ecosystem became more diverse, even as Bitcoin remained dominant.

Despite various setbacks, such as the 2016 Bitfinex hack (approximately BTC 120,000 stolen), the market remained resilient, absorbing the shock without systemic collapse. During this phase, Bitcoin's price, which had bottomed out below USD 200 in early 2015, steadily recovered toward USD 1,000 by the end of 2016. At the same time, rolling volatility declined from peaks above 200 to 300% in 2014 to levels near 50 to 60%. The 2016 halving, which reduced the block reward from BTC 25 to 12.5, further entrenched the programmed-scarcity narrative and reinforced Bitcoin's long-term valuation model. Together, these developments marked the regime's stabilization, declining risk premia, and the first concrete steps toward integration with traditional finance.

The third phase, spanning from late 2016 through 2024, encompasses Bitcoin's most intense expansionary and contractionary cycles to date. This period includes the 2017 bull market and subsequent crash, which were fueled by retail speculation and the boom in initial coin offerings. Significantly, December 2017 also marked the launch of Bitcoin futures contracts on the CME and CBOE, signaling Bitcoin's initial integration into established financial markets and facilitating basis trading and arbitrage [Kapar and Olmo, 2019, Makarov and Schoar, 2020, 2021]. In parallel, the explosive growth of perpetual futures—initially popularized by BitMEX and scaled globally through Binance Futures from 2019 onward—reshaped price discovery by introducing a funding-driven leverage cycle.

Imbalances in funding rates became closely associated with liquidation cascades and

²In the structural break analysis based on trading volume, the identified break-points proved to be highly consistent with those obtained from the price-based specification.

short-squeeze dynamics. The rise of Binance (founded in 2017) as the dominant spot and derivatives venue amplified these effects by concentrating liquidity and enabling continuous cross-margining across spot, futures, and perpetual futures. In parallel, the emergence of decentralized exchanges (DEXs)—from early order-book models such as EtherDelta in 2017 to automated market makers like Uniswap in 2018—broadened the available venues and introduced on-chain liquidity that interacted with centralized markets, particularly during stress episodes and catalytic events [Adams et al., 2021, Angeris et al., 2022, Werner et al., 2022]. After the deep bear market from 2018 to 2019, Bitcoin’s recovery was hastened by the unprecedented monetary policy response to the COVID-19 pandemic, including near-zero interest rates and large-scale quantitative easing. These conditions drove Bitcoin to new all-time highs; it reached above USD 60,000 in 2021. Overall, this regime illustrates Bitcoin’s transformation from a niche speculative instrument into a macro-sensitive, high-beta risk asset increasingly correlated with traditional financial variables.

The fourth regime considered here began in 2024 with the onset of greater institutional adoption and full-scale financial integration. The approval and rapid growth of U.S. spot Bitcoin ETFs, launched by prominent asset managers such as BlackRock, Fidelity, and Ark, was a watershed moment, with traditional investors gaining regulated access to Bitcoin through familiar financial vehicles. Ethereum spot ETFs were introduced shortly thereafter, further expanding the investable crypto universe and strengthening the perception of digital assets as a permanent asset class. Parallel developments in Europe and Canada, where crypto exchange-traded products had already gained traction, consolidated a global market for exchange-traded crypto products.

Institutional participation was reinforced by corporate treasury allocations, with MicroStrategy surpassing BTC 200,000 on its balance sheet and other companies, such as Tesla, continuing to hold Bitcoin as part of their financial reserves. The concept of “Bitcoin treasury companies” also became more prominent, signaling the asset’s integration into corporate finance strategies. At the regulatory level, the EU’s Markets in Crypto-Assets (MiCA) regulatory framework and parallel initiatives in Asia provided greater legal certainty for institutional allocators, reducing compliance barriers and enhancing the assets’ legitimacy.

In this period, Bitcoin’s institutionalization accelerated. No longer confined to speculative trading on the fringes of finance, Bitcoin and Ethereum became embedded in the broader portfolio-allocation decisions of pension funds, wealth managers, and other institutional investors. This process is evident in the rapid expansion of CME futures and options volumes, with Bitcoin now traded through the same derivative infrastructure as traditional assets. These developments signal a structural shift: cryptocurrencies are no longer niche instruments but investable assets in the global financial architecture.

Overall, the identified structural breaks align closely with major milestones in Bitcoin’s

evolution: the collapse of the early infrastructure (2014), the rise of robust exchanges and the programmed-scarcity narrative (2016), the expansion of derivatives markets, perpetuals, Binance’s dominance, and the emergence of DEXs (2016–2024), as well as the formal institutionalization via spot ETFs and regulatory clarity (2024 onwards). These findings suggest that Bitcoin’s statistical regimes are not entirely endogenous to its price dynamics but correspond to broader shifts in market structure, trading technology, and macro-financial integration.

The sparse CP model identifies broad statistical regimes based on shifts in Bitcoin’s return dynamics. For the causality analysis in Section 5, we partition the 2016 to 2024 regime into shorter, event-defined subperiods—bull-run era (2017–2018), institutional adoption (2019–2021), and bear market (2022–2023)—to avoid averaging across distinct market structures. The 2017 to 2018 subperiod captures the retail-driven boom in initial coin offerings and initial derivative launches, 2019 to 2021 covers the COVID-19 liquidity regime and unprecedented monetary expansion, and the period from 2022 to 2023 reflects monetary tightening and the deleveraging triggered by major market dislocations. We treat these as sub-regimes within the broader CP regime rather than separate structural breaks. This approach allows us to examine how predictability patterns vary across market conditions while remaining consistent with the statistically identified regime structure.

The rolling volatility analysis shows a pronounced decline over time, underscoring the progressive maturation of the Bitcoin market. During its formative phase, annualized volatility frequently exceeded 200% to 300%, a magnitude typical of an experimental digital asset during initial market discovery and institutional framing [Baur et al., 2018]. The first structural break around 2014 to 2015 coincides with a regime shift: the persistent reduction in volatility suggests a deepening market structure, which is characterized by the emergence of early institutional players, the consolidation of trading venues, and the development of arbitrage mechanisms across platforms.



Figure 2: Bitcoin structural regimes (2010–2014, 2014–2016, 2016–2024, 2024–present).

In 2020, this downward trajectory was temporarily overshadowed by a spike in volatility during the COVID-19 shock. As was the case for other financial markets, Bitcoin was affected by the sudden flight-to-liquidity in March 2020, with annualized volatility soaring to levels reminiscent of its early trading years. This episode illustrates that—even when considered a potential hedge—Bitcoin remains vulnerable to systemic global liq-

uidity stress [Corbet et al., 2020].

Following this episode, the trend stabilized, with volatility converging to lower levels. This development coincides with the increased entry of institutional investors from 2020 and 2021 onwards, including listed funds, regulated custodians, and, more recently, spot Bitcoin ETFs. Institutional participation tends to dampen volatility through hedging, increased liquidity, and cross-asset arbitrage [Akyildirim et al., 2023]. The figure reinforces the narrative that Bitcoin is evolving from a highly speculative asset into a more integrated component of the global financial system. Recent empirical evidence reinforces this interpretation of the pattern. Babalos et al. [2025] report lower volatility for both Bitcoin and Ripple after the launch of U.S. spot ETFs, suggesting that the introduction of regulated investment vehicles has helped stabilize the market. These findings highlight the role of institutional mechanisms in consolidating the maturity of the Bitcoin market.

2.3. Literature Review

2.3.1. Bibliometric Review

Bibliometric evidence from Scopus reveals that the literature on Bitcoin has evolved through distinct phases aligned with market cycles. Early work (2014–2016) was driven by computer science, emphasizing privacy mechanisms, smart contracts, and consensus protocols. During the 2017 to 2018 bull run, research pivoted toward scalable architecture and permissioned blockchain systems. From 2019 onwards, the literature broadened in volume and disciplinary scope to include domain-specific applications (healthcare, supply chains, IoT) and financial economics. This latter stream increasingly focused on uncertainty quantification, cross-market connectedness, and spillovers between cryptocurrencies and traditional assets.

Table 1: Top 5 most cited papers by period (Scopus, keyword “Bitcoin”)

Title	Year	Citations	Source title
<i>Early Bitcoin (2014–2016)</i>			
Decentralizing Privacy: Using Blockchain to Protect Personal Data	2015	2186	IEEE Symposium on Security and Privacy Workshops (SPW)
Making smart contracts smarter	2016	1866	Proceedings of the ACM Conference on Computer and Communications Security
Where is current research on Blockchain technology? – A systematic review	2016	1803	PLOS ONE
Bitcoin and beyond: A technical survey on decentralized digital currencies	2016	1398	IEEE Communications Surveys and Tutorials
On the security and performance of Proof of Work blockchains	2016	1358	Proceedings of the ACM Conference on Computer and Communications Security
<i>Bull Run Era (2017–2018)</i>			
An Overview of Blockchain Technology: Architecture, Consensus, and Future Trends	2017	3658	IEEE International Conference on Services Economics (SE)
Hyperledger Fabric: A Distributed Operating System for Permissioned Blockchains	2018	2776	EuroSys (European Conference on Computer Systems)
IoT security: Review, blockchain solutions, and open challenges	2018	2070	Future Generation Computer Systems
Algorand: Scaling Byzantine Agreements for Cryptocurrencies	2017	1279	SOSP: ACM Symposium on Operating Systems Principles
Ouroboros: A provably secure proof-of-stake blockchain protocol	2017	1094	Lecture Notes in Computer Science
<i>Institutional Adoption (2019–2021)</i>			

Table 1 (continued)

Title	Year	Citations	Source title
Blockchain technology in healthcare: A systematic review	2019	853	Healthcare (Switzerland)
Applications of Blockchains in the Internet of Things: A Comprehensive Survey	2019	744	IEEE Communications Surveys and Tutorials
A Survey of Distributed Consensus Protocols for Blockchain Networks	2020	730	IEEE Communications Surveys and Tutorials
Blockchain technology for enhancing supply chain resilience	2019	709	Business Horizons
The contagion effects of the COVID-19 pandemic: Evidence from gold and cryptocurrencies	2020	676	Finance Research Letters
<i>Bear Market (2022–2023)</i>			
Blockchain Technology: Benefits, Challenges, Applications, and Integration of Blockchain Technology with Cloud Computing	2022	283	Future Internet
The cryptocurrency uncertainty index	2022	233	Finance Research Letters
Emerging Trends in Blockchain Technology and Applications: A Review and Outlook	2022	229	Journal of King Saud University – Computer and Information Sciences
Does oil connect differently with prominent assets during war? Analysis of intra-day data during the Russia–Ukraine saga	2022	223	Resources Policy
Static and dynamic connectedness between NFTs, DeFi and other assets: Portfolio implication	2022	211	Global Finance Journal
<i>Recovery (2024)</i>			
Impact of Artificial Intelligence and Cyber Security as Advanced Technologies on Bitcoin Industries	2024	107	International Journal of Intelligent Systems and Applications in Engineering
Cryptocurrency awareness, acceptance, and adoption: the role of trust as a cornerstone	2024	54	Humanities and Social Sciences Communications

Table 1 (continued)

Title	Year	Citations	Source title
Forecasting Bitcoin prices using artificial intelligence: Combination of ML, SARIMA, and Facebook Prophet models	2024	49	Technological Forecasting and Social Change
Toward a greener future: A survey on sustainable blockchain applications and impact	2024	47	Journal of Environmental Management
A Comprehensive Study of Artificial Intelligence and Cybersecurity on Bitcoin, Crypto Currency and Banking System	2024	45	Annals of Data Science

Note. “Citations” are Scopus Cited.by counts at extraction time. Source titles reflect your venue corrections where provided.

Figure 3 summarizes the evolution of the on-chain subset under our classification. Explicit on-chain analytics papers are rare in the early years of our sample. There are more of these after 2017, but they remain disproportionately concentrated in computer-science outlets relative to finance or economics venues.

In our Scopus corpus ($N = 13,515$), papers classified as using on-chain data represent 1.3% ($N = 177$). These studies are primarily published in computer science and engineering outlets (e.g., IEEE: 46; Springer: 34 in this snapshot), with comparatively limited presence in finance journals where asset pricing, portfolio allocation, and systemic risk are central questions. The temporal pattern is consistent with a gradual diffusion of on-chain methods into economic applications: the on-chain subset is small in 2014–2016, grows slowly during 2017–2018, and increases more noticeably after 2019. Work using on-chain metrics explicitly for return prediction and valuation becomes more common during 2022–2023, coinciding with heightened interest in crypto-market stress and leverage dynamics.

The concentration described above suggests a methodological gap. In much of the financial economics literature, Bitcoin is studied using price-based time series (e.g., volatility transmission and spillovers). However, despite the transparency of the underlying ledger, it typically does not incorporate blockchain-native fundamentals. On-chain data—such as transaction activity, active addresses, hash rate, measures based on unspent transaction output (UTXO), and exchange flows—offers a complementary view of network usage and negotiable supply that does not rely on exchange-mediated proxies.

Given the small share of on-chain-focused papers in the overall corpus, integrating these measures into mainstream econometric connectedness and spillover designs remains uncommon. A plausible explanation for this gap is disciplinary specialization. For instance, computer-science researchers often have the data infrastructure and protocol expertise to construct on-chain measures, while finance research usually emphasizes econometric identification and cross-asset modeling. A desire to bridge these approaches motivates our empirical design, which embeds on-chain indicators into causality frameworks alongside macro-financial variables [Hoang and Baur, 2022, Baur and Hoang, 2021].

2.3.2. Evolution of On-Chain Metrics and Literature Evidence

On-chain data refers to the information directly recorded on public blockchains, such as Bitcoin, encompassing every transaction, block, and address observable in real time. Unlike market-based data sourced from centralized exchanges—prices, trading volumes, or order books—on-chain data represents the native, verifiable layer of activity within the distributed ledger itself. This transparency enables the construction of novel indicators of value, liquidity, adoption, and investor behavior, although methodologi-

The On-Chain Research Gap in Bitcoin Literature

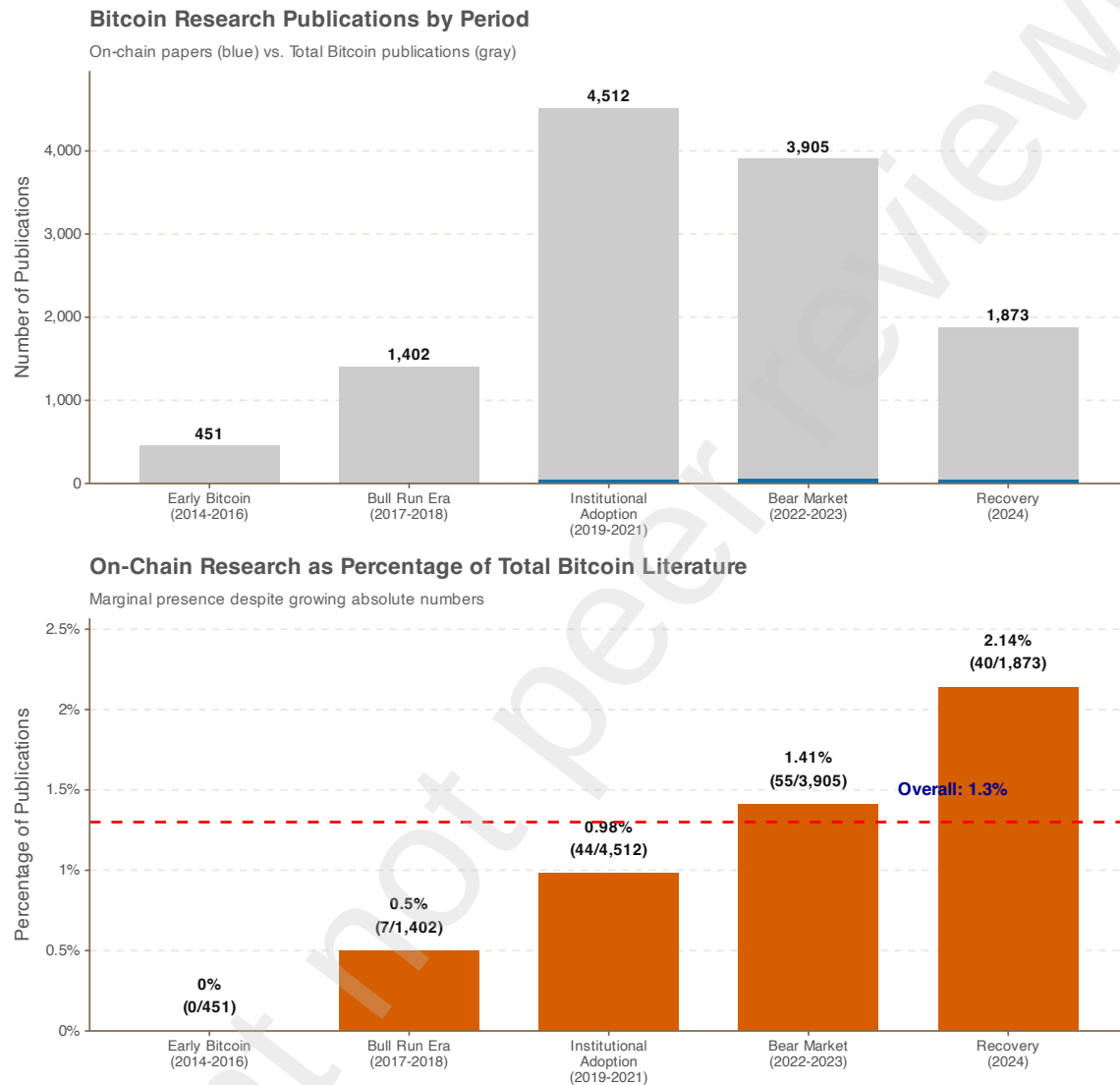


Figure 3: The On-Chain Research Gap in Bitcoin Literature

Notes: **Panel A** shows total Bitcoin publications (gray) and on-chain subset (blue) by period. **Panel B** displays on-chain research as percentage of period total, with dashed line indicating overall 1.3% representation. Despite accelerating growth post-2019, on-chain analysis remains marginal, never exceeding 1.2% of publications within any period. This gap is particularly striking given blockchain transparency: every transaction is publicly observable, yet academic research overwhelmingly relies on exchange-mediated market data rather than blockchain-native fundamentals. Data source: Scopus; $N_{\text{total}} = 13,515$, $N_{\text{on-chain}} = 177$.

cal challenges remain, such as distinguishing genuine economic transfers from internal reshuffling by custodians.

2.3.3. On-Chain Metrics: Concept and Evolution

On-chain data are recorded directly on a public blockchain. For Bitcoin, this includes transactions, blocks, and address-level activity that can be observed as the ledger updates. In contrast to market data obtained from centralized exchanges (prices, volumes, or order-book variables), on-chain series describe activity in the settlement layer. This transparency supports the construction of measures of adoption, liquidity conditions, and investor behavior. At the same time, interpretation is not mechanical. A recurrent difficulty is separating economically meaningful transfers from operational flows, such as custodial rebalancing and internal “self-churn.”

The use of on-chain indicators has also evolved in stages. Early studies relied on basic counts, including transaction totals, active addresses, and aggregate transfer volume [Nakamoto, 2008, Ron and Shamir, 2013]. These variables were informative as broad proxies for usage, but were often noisy because they are a mix of genuine payment activity and internal movements by exchanges and service providers.

As the ecosystem has matured, researchers have developed entity-adjusted measures that attempt to map addresses to economic actors. Meiklejohn et al. [2013] and Harrigan and Fretter [2016] show how clustering heuristics, such as multi-input rules and change-address detection, can approximate entities and reduce the influence of internal accounting operations. This shift improved the measurement of net flows between participants and helped link ledger activity to economic behavior [Kalodner et al., 2017].

In a second strand of innovation, value-based indicators derived from the UTXO set were introduced. Realized capitalization, for example, values each coin at the price at which it last moved rather than at the current market price. The MVRV became a widely used summary statistic in practitioner research and is increasingly used in academic work on speculative cycles and valuation dynamics [Glassnode, 2020, Kraussl et al., 2020].

Related measures focus on realized gains and losses. The spent output profit ratio indicates whether coins that move on a given day are, on average, spent at a gain or a loss. Measures of coin aging, such as “coin days destroyed”, provide complementary information by distinguishing periods of long-term holding from distribution episodes [Baur and Hoang, 2021].

Exchange flow variables have become central in the literature on liquidity and short-horizon price pressure. Hoang and Baur [2022] emphasize exchange reserves and net

inflows or outflows as indicators of negotiable supply and potential stress in intermediated markets. Early informal discussions of exchange solvency in Bitcoin markets were later formalized in the academic literature. In particular, Dagher et al. [2015]) propose Provisions, a cryptographic protocol that allows centralized exchanges to prove solvency by jointly demonstrating ownership of on-chain reserves and sufficient coverage of customer liabilities, while preserving user privacy through zero-knowledge techniques. This framework laid the foundation for modern proof-of-reserves and proof-of-solvency mechanisms and remains the canonical academic reference in this area. Broader surveys of Bitcoin security and market infrastructure, such as Bonneau et al. [2015], further contextualize proofs of solvency as a response to custodial risk in centralized intermediaries.

New uses of blockspace have complicated the interpretation of standard activity metrics. The introduction of ordinals and BRC-20 tokens [Rodarmor, 2023] changed fee dynamics and increased competition for inclusion in blocks. As discussed in recent work, a growing share of the observed activity may reflect non-monetary use cases, making transaction-based indicators less directly comparable to those in earlier periods [Bertucci, 2024, Cerutti et al., 2024].

2.3.4. Applications in the Literature

Although the measurement toolkit has expanded rapidly, there remain relatively few empirical applications of on-chain variables in finance, and the evidence is scattered across topics. Existing studies nevertheless illustrate both the potential of ledger-based measures and the constraints imposed by attribution and measurement error.

Cerutti et al. [2024] use on-chain transactions to study cross-border Bitcoin flows and show that this activity responds to macroeconomic shocks and capital controls. Their results complement balance-of-payments statistics and highlight the limits of geographic attribution. In a market microstructure setting, Hoang and Baur [2022] examine exchange balances and find that net outflows from exchanges to private wallets are associated with higher subsequent returns. In contrast, net inflows tend to precede selling pressure. This evidence positions exchange reserves and net flows as short-horizon liquidity indicators.

Other contributions focus on supply dynamics and timing. Ambrosia et al. [2024] document changes in Bitcoin's active supply that are consistent with long-horizon holder accumulation and a declining negotiable float, with effects that intensified around the 2024 halving. Jahanshahloo et al. [2022] show that on-chain activity follows pronounced intraday patterns that coincide with the opening of U.S. equity markets, suggesting that some measures of adoption and usage may co-move with global trading schedules. On-chain methods have also been used outside asset pricing. Amiram et al. [2022] analyze transaction patterns associated with illicit finance and argue that unusual flows

can contain advanced information about security-related events, underscoring their regulatory and compliance relevance.

Overall, these studies point to a gap in the literature. On-chain indicators are well-measured but are rarely integrated into standard empirical finance frameworks alongside macro-financial variables. Our paper addresses this by embedding blockchain-native measures and market variables in a unified causality framework and by examining how their predictive relationships change across regimes.

3. Methodology

Our analysis focuses on two parts: the daily relationship between Bitcoin and on-chain and financial market variables, such as Gold, VIX, WTI, NASDAQ, and DXY, as well as monetary variables, such as the yield curve slope, the 10-year Treasury, and the U.S. money supply (M2). We use linear and nonlinear analysis to understand how these relationships can be made. We also analyze monthly data, since M2 is available only monthly. To this end, we apply the Breitung and Candelon [2006] approach, which can disentangle short and long-run relationships.

The analysis of causal relationships in financial time series has traditionally relied on time-domain methods. However, these fail to capture the frequency-specific nature of causality, potentially missing essential dynamics that operate at different time scales. The Breitung–Candelon frequency-domain approach addresses this limitation by decomposing causality across the frequency spectrum.

3.1. Frequency-Domain Granger Causality

We study directional predictability across frequencies using frequency-domain (spectral) Granger causality [Geweke, 1982] and the Breitung–Candelon (BC) test [Breitung and Candelon, 2006].

Let x_t and y_t be (weakly) stationary scalar processes collected in $\mathbf{z}_t = (y_t, x_t)'$ that follow a stable VAR(p):

$$\mathbf{z}_t = \sum_{j=1}^p \Phi_j \mathbf{z}_{t-j} + \boldsymbol{\varepsilon}_t, \quad \mathbb{E}[\boldsymbol{\varepsilon}_t] = \mathbf{0}, \quad \mathbb{E}[\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t'] = \Sigma. \quad (3)$$

Define the lag polynomial $\Phi(z) = \sum_{j=1}^p \Phi_j z^j$ and the transfer function $\Psi(z) = (\mathbf{I} - \Phi(z))^{-1}$. The spectral density of $\mathbf{z}_t$ at frequency $\omega \in (0, \pi]$ is

$$\mathbf{f}(\omega) = \frac{1}{2\pi} \Psi(e^{-i\omega}) \Sigma \Psi(e^{i\omega})'. \quad (4)$$

Thus, x_t Granger-causes y_t at frequency ω if the component of y_t at ω is improved by including the past of x_t . Geweke's measure can be written as

$$\mathcal{F}_{x \rightarrow y}(\omega) = \ln \left(\frac{S_{yy}(\omega)}{S_{yy|x}(\omega)} \right) \geq 0, \quad (5)$$

where $S_{yy}(\omega)$ is the $(1, 1)$ element of $\mathbf{f}(\omega)$ and $S_{yy|x}(\omega)$ is the spectrum when the x -channel is shut down.

Breitung and Candelon [2006] show that testing “no causality from x to y at ω ” is equivalent to testing two linear restrictions on the coefficients of the x -lags in the y -equation. Let $\phi_{yx,j}$ denote the coefficient on x_{t-j} in the y_t equation, $j = 1, \dots, p$. Define

$$\alpha(\omega) = \sum_{j=1}^p \phi_{yx,j} \cos(j\omega), \quad \beta(\omega) = \sum_{j=1}^p \phi_{yx,j} \sin(j\omega). \quad (6)$$

Then the null at frequency ω is

$$H_0(\omega) : \quad \alpha(\omega) = 0 \text{ and } \beta(\omega) = 0. \quad (7)$$

Let $\hat{\beta}$ stack the OLS coefficients of the y_t equation in (3) and let $\widehat{\mathbf{V}} = \widehat{\text{Var}}(\hat{\beta})$. If $\mathbf{R}(\omega)$ selects the $\phi_{yx,j}$ and premultiplies them by $\{\cos(j\omega), \sin(j\omega)\}$ as in (7), the Wald statistic is

$$\mathcal{W}(\omega) = T \left(\mathbf{R}(\omega) \hat{\beta} \right)' \left[\mathbf{R}(\omega) \widehat{\mathbf{V}} \mathbf{R}(\omega)' \right]^{-1} \left(\mathbf{R}(\omega) \hat{\beta} \right) \xrightarrow{d} \chi^2(2), \quad (8)$$

under $H_0(\omega)$ and standard regularity conditions. Rejection implies frequency-specific Granger causality $x \rightarrow y$ at ω . Because tests are performed over many ω , control size using Bonferroni ($\alpha^* = \alpha/M$) or, preferably, Benjamini–Hochberg FDR at level q .

3.2. Transfer Entropy

We quantify directional dependence between time series by employing TE, an information-theoretic measure of the predictive information that one stochastic process conveys about the future of another [Schreiber, 2000]. TE is nonparametric and captures potentially nonlinear, asymmetric relationships, in contrast to traditional parametric approaches (e.g., cointegration tests or multivariate generalized autoregressive conditional heteroskedasticity models), which primarily accommodate linear dependence structures.

Let X be a discrete random variable with probability mass function $p(x)$. The Shannon entropy is

$$H(X) = - \sum_x p(x) \log p(x), \quad (9)$$

measuring average uncertainty (we use natural logarithms; results are in *nats*). For two variables X and Y with joint pmf $p_{XY}(x, y)$ and marginals $p_X(x)$, $p_Y(y)$, the mutual information (MI) is

$$I(X; Y) = \sum_{x, y} p_{XY}(x, y) \log \frac{p_{XY}(x, y)}{p_X(x) p_Y(y)}, \quad (10)$$

which equals the Kullback–Leibler divergence between the joint and the product of marginals. MI is nonnegative and equals zero if and only if X and Y are independent.

Consider two stationary processes $\{X_t\}$ and $\{Y_t\}$. Let $X_t^{(k)} = (X_t, \dots, X_{t-k+1})$ and $Y_t^{(l)} = (Y_t, \dots, Y_{t-l+1})$ denote k - and l -dimensional past histories, respectively. Transfer entropy from Y to X is the conditional mutual information

$$T_{Y \rightarrow X}(k, l) = I(X_{t+1}; Y_t^{(l)} | X_t^{(k)}) = \sum p(x_{t+1}, x_t^{(k)}, y_t^{(l)}) \log \frac{p(x_{t+1} | x_t^{(k)}, y_t^{(l)})}{p(x_{t+1} | x_t^{(k)})}. \quad (11)$$

TE is nonnegative and equals zero if and only if $p(X_{t+1} | X_t^{(k)}, Y_t^{(l)}) = p(X_{t+1} | X_t^{(k)})$ almost surely, i.e., Y provides no incremental predictive content for X beyond X 's own past. Note that, in general, $T_{Y \rightarrow X} \neq T_{X \rightarrow Y}$, so TE is inherently directional.

Following Dimpfl and Peter [2014], we estimate TE using a discrete (symbolic) representation of returns. Let r_t denote the return series and let $q_{0.05}$ and $q_{0.95}$ be the 5th and 95th sample quantiles of $\{r_t\}$. We recode each observation into three symbols,

$$S_t = \begin{cases} A, & \text{if } r_t \leq q_{0.05}, \\ B, & \text{if } q_{0.05} < r_t < q_{0.95}, \\ C, & \text{if } r_t \geq q_{0.95}, \end{cases} \quad (12)$$

which capture large negative moves, typical fluctuations, and large positive moves, respectively. With (S_t^X, S_t^Y) denoting the symbolic versions of (X_t, Y_t) , we form empirical joint and conditional frequencies to plug into (11). Embedding dimensions k, l are set to small integers (e.g., $k = l \in \{1, 2\}$) and can be chosen by information criteria or out-of-sample performance.

We assess significance using a permutation (or stationary block bootstrap) null that destroys cross-series linkages while preserving serial dependence within each series: we repeatedly permute (or block-shuffle) $\{Y_t\}$, recompute $T_{Y \rightarrow X}^*$, and form empirical p -values from the reference distribution.

4. Data

The dataset combines on-chain metrics and market variables over the period 2013–2025. The on-chain variables, obtained from CryptoQuant ³, comprise up to 4,474 daily observations covering the period from April 29, 2013, to July 28, 2025. These data series include measures of whale activity, exchange reserves and netflows, active addresses, the hash rate, transaction fees, funding rates, open interest, and the MVRV ratio.

Data for market variables are obtained from Thomson Reuters (Refinitiv), with 3,035 trading-day readings on the New York Stock Exchange from April 30, 2013, to July 30, 2025. The variables include the VIX, DXY, Gold, NASDAQ Composite, and WTI crude. Monetary and rate series come from the Federal Reserve Economic Data program, yielding 145 to 146 monthly observations (June 2013 to July 2025) for M2, the 10Y–2Y yield curve slope, and the 2Y and 10Y Treasury yields. Everything was differenced or otherwise transformed to make it stationary, then lined up at daily or monthly intervals to run frequency-domain causality tests.

³See the methodology for on-chain data at <https://userguide.cryptoquant.com/>

Table 2: Description of variables used in the analysis

Category	Variable	Obs	Description / Formula	Source
On-chain	Whale Activity	4474	Variation in positions of large holders (whales)	CryptoQuant
	Exchange Reserves	4474	Aggregate BTC balances on centralized exchanges	CryptoQuant
	Exchange Netflow	4474	Net inflows to exchanges: $\text{Netflow}_t = \text{In}_t - \text{Out}_t$	CryptoQuant
	Active Addresses	4474	Number of unique active Bitcoin addresses	CryptoQuant
	Hash Rate	4474	Network computational power (EH/s)	CryptoQuant
	Transaction Fees	4474	Total transaction fees paid (proxy for congestion)	CryptoQuant
	Funding Rate	3363	Perpetual futures funding rate: $\text{FR}_t = (\frac{P_{fut}}{P_{spot}} - 1)$ adj. by period	CryptoQuant
	Open Interest	2314	Open interest in futures/perpetual contracts	CryptoQuant
	MVRV Ratio	4474	Market to realized value ratio: $\text{MVRV}_t = \frac{\text{Market Cap}_t}{\text{Realized Cap}_t}$	CryptoQuant
Market	VIX	3035	Implied volatility index (S&P 500 options)	Thomson Reuters
	DXY	3035	Dollar index returns: $r_t^{DXY} = \ln(P_t) - \ln(P_{t-1})$	Thomson Reuters
	Gold	3033	Gold returns: $r_t^{Gold} = \ln(P_t) - \ln(P_{t-1})$	Thomson Reuters
	Nasdaq	3035	Nasdaq Composite returns: $r_t^{Nasdaq} = \ln(P_t) - \ln(P_{t-1})$	Thomson Reuters
	WTI	3035	WTI crude oil returns: $r_t^{WTI} = \ln(P_t) - \ln(P_{t-1})$	Thomson Reuters
Monetary	Yield Curve Slope	146	Term structure: $\text{Slope}_t = y_t^{10Y} - y_t^{2Y}$	U.S. Treasury
	Treasury 10Y	146	10-year nominal yield (y_t^{10Y})	U.S. Treasury
	Treasury 2Y	146	2-year nominal yield (y_t^{2Y})	U.S. Treasury
	M2	145	Broad money supply (M2, seasonally adjusted)	FRED

On-chain frequency: daily (calendar days, 2013-04-29 to 2025-07-28; $N=4474$).

Market frequency: NYSE trading days (2013-04-30 to 2025-07-30; $N=3035$).

Monetary frequency: monthly (2013-06 to 2025-07; $N=145-146$).

All variables were log-transformed or differenced for stationarity before analysis.

5. Results

This section presents the empirical findings from two complementary causality frameworks: frequency-domain Granger causality tests [Breitung and Candelon, 2006] and TE analysis. The linear approach captures predictive relationships at specific cyclical frequencies, while the nonlinear framework detects information flows arising from regime-dependent dynamics, threshold effects, or asymmetric responses. Throughout our analysis, “causality” is used in the Granger/information-theoretic sense of predictive content, not in the structural-causation sense. By examining both methodologies across five distinct market phases—early Bitcoin (2014–2016), bull run era (2017–2018), institutional adoption (2019–2021), bear market (2022–2023), and recovery (2024)—we document a marked reconfiguration in the predictive linkages between Bitcoin and global financial markets.

5.1. Early Bitcoin (2014–2016): Network-Dominated Dynamics

During Bitcoin’s formative phase, the market was less connected to traditional financial assets than in later regimes, with macro predictability mostly confined to very short horizons. Linear causality tests (Table 3) show that several macro variables predict Bitcoin at short cycles: NASDAQ ($\chi^2 = 11.20$, $p < 0.01$), WTI Oil ($\chi^2 = 10.99$, $p < 0.01$), VIX ($\chi^2 = 9.04$, $p < 0.05$), Term Structure ($\chi^2 = 7.17$, $p < 0.05$), and 10-Year Treasury ($\chi^2 = 5.66$, $p < 0.10$) are all significant at cycles of approximately 2 to 3 days. These χ^2 statistics are generally smaller than those of key on-chain variables in the same period. Reverse predictability is limited but not absent. BTC→WTI is significant at the 1% level, and smaller effects appear for BTC→DXY and BTC→Term Structure at the 10% level. This asymmetry—stronger inbound than outbound macro linkages—aligns with an emerging, thinly traded market with modest spillovers to traditional assets.

On-chain variables contain comparatively stronger lead-lag predictability for Bitcoin returns during this period. Exchange Reserves ($\chi^2 = 16.69$, $p < 0.01$) and Exchange Netflow ($\chi^2 = 13.07$, $p < 0.01$) exhibit significant effects at short frequencies (≈ 3 days), while the MVRV Ratio has predictability at longer cycles (34.3 days). These patterns indicate that, in this phase, crypto-native supply and liquidity measures have stronger predictive content than macro variables.

The TE results (Table 8) are generally insignificant for most on-chain and macro variables in this phase, with several notable exceptions. On the macro side, 10-year Treasury→BTC is significant (TE = 0.0187, $p < 0.01$), which is early evidence of Bitcoin’s sensitivity to interest rate conditions through nonlinear channels. In the reverse direction, BTC→NASDAQ (TE = 0.0193, $p < 0.01$) and BTC→MVRV (TE = 0.0141, $p < 0.01$) are significant, suggesting that Bitcoin returns contained forecasting content

for the NASDAQ index and that this valuation operates through nonlinear pathways. The comparison of linear and nonlinear results during this period identifies stronger on-chain predictors, while TE detects only a small number of nonlinear channels (notably $10Y \rightarrow BTC$, $BTC \rightarrow NASDAQ$, and $BTC \rightarrow MVRV$). This pattern is consistent with a market that has an information structure that is predominantly endogenous and approximately linear.

5.2. Bull Run Era (2017–2018): Emerging Financial Integration

The 2017–2018 period marks the onset of Bitcoin’s integration into broader financial markets. Linear causality tests (Table 4) show that connections with traditional assets remained modest. NASDAQ and Term Structure are marginally significant ($p < 0.10$) at medium-term cycles (≈ 15 – 16 days), while Gold is marginally significant at a shorter cycle (≈ 3.5 days). Reverse causality strengthened notably: Bitcoin began transmitting information to the VIX ($p < 0.10$), USD ($p < 0.10$), NASDAQ ($p < 0.10$), Term Structure ($p < 0.05$), and 10-Year Treasury ($p < 0.05$).

On-chain metrics retained substantial predictive power. The MVRV Ratio emerges as the dominant linear predictor ($\chi^2 = 72.05$, $p < 0.01$), with causality concentrated at short cycles (3.4 days). Exchange Reserves ($\chi^2 = 16.83$, $p < 0.01$) and Exchange Netflow ($\chi^2 = 13.03$, $p < 0.01$) also have significant effects. Active Addresses has predictive content at very low frequency (628.3 days, near the lower-frequency bound in our frequency grid rather than a literal two-year cycle); this is consistent with slow-moving network-adoption effects rather than short-horizon trading pressure. Bidirectional predictability between BTC and Funding Rates is consistent with a larger role for derivatives positioning during this period.

The nonlinear analysis reveals additional structure. TE identifies a significant information flow to BTC from Hash Rate (TE = 0.0220, $p < 0.01$) and Exchange Netflow (TE = 0.0212, $p < 0.01$). These links are also significant in the linear tests, and the TE evidence indicates that they persist when dependence is measured using an information-theoretic metric that is sensitive to nonlinear structure. Most notably, $MVRV \rightarrow BTC$ exhibits strong nonlinear predictability (TE = 0.0755, $p < 0.01$), the largest TE effect in this regime. In the reverse direction, Whale Activity (TE = 0.0285, $p < 0.01$) and Funding Rate (TE = 0.0192, $p < 0.01$) exhibit significant nonlinear responses to Bitcoin returns, suggesting that large holders and derivatives traders make threshold- or state-dependent adjustments in response to price movements.

The divergence between the linear and nonlinear results during the bull-run phase indicates nonlinear characteristics emerging in Bitcoin’s relationship with network fundamentals; a plausible interpretation of this is that market participants responded asymmetrically during boom and bust phases of the speculative cycle.

5.3. Institutional Adoption (2019–2021): Macro-Financial Integration

During the 2019 to 2021 period, there is a strengthening of macro-financial predictability. In the linear causality tests (Table 5) VIX ($\chi^2 = 32.26$), USD ($\chi^2 = 34.35$), Gold ($\chi^2 = 14.87$), and NASDAQ ($\chi^2 = 27.93$) have a significant causal relationship with Bitcoin (all $p < 0.01$), with effects concentrated at short-to-medium frequencies. The COVID-19 pandemic occurred during this period, alongside unprecedented monetary expansion and increased institutional participation in cryptocurrency markets.

On-chain and derivatives metrics exhibited even stronger forecasting power. The MVRV Ratio achieves its highest linear test statistic across all periods ($\chi^2 = 604.30$, $p < 0.01$), while Open Interest ($\chi^2 = 145.98$, $p < 0.01$) and Funding Rate ($\chi^2 = 52.06$, $p < 0.01$) are revealed to be powerful predictors at medium-term cycles (roughly 21–32 days). These results align with the maturation of the derivatives infrastructure and an increasing role for leverage in Bitcoin’s price-discovery process.

Nonlinear evidence reinforces and extends these findings. TE from MVRV Ratio (TE = 0.0909, $p < 0.01$) and Open Interest (TE = 0.0549, $p < 0.01$) to BTC is highly significant, indicating the flow of predictive information beyond linear channels. The 10-Year Treasury yield continues to have a nonlinear causal relationship with BTC (TE = 0.0219, $p < 0.01$), while there is significant nonlinear information transmission from the Hash Rate (TE = 0.0178, $p < 0.01$).

By contrast, macro-financial variables exhibit limited nonlinear causality despite their strong linear effects. The variables VIX, DXY, Gold, and NASDAQ all lack significant TE toward Bitcoin in this regime (the 10-year Treasury is the exception). This asymmetry suggests that Bitcoin’s sensitivity to variables such as the VIX and DXY operates primarily through contemporaneous or near-contemporaneous linear channels. At the same time, on-chain and derivatives metrics transmit information through linear and nonlinear pathways. The pattern is consistent with the possibility that network fundamentals condition Bitcoin’s response to macro conditions in a state-dependent manner, though this interpretation requires further investigation.

5.4. Bear Market (2022–2023): Amplified Macro Sensitivity

The 2022 to 2023 monetary tightening cycle produced the strongest macro-financial linear predictability in our sample. Linear causality tests (Table 6) show that VIX ($\chi^2 = 54.00$), USD ($\chi^2 = 27.98$), and NASDAQ ($\chi^2 = 98.94$) have highly significant effects on Bitcoin (all $p < 0.01$). The 10-year Treasury yield ($\chi^2 = 16.00$, $p < 0.01$) also has predictive power at longer cycles (27.3 days), in line with greater sensitivity to interest-rate conditions during the Federal Reserve’s rate-hike campaign.

A notable shift occurred in on-chain dynamics. Exchange Reserves and Exchange

Netflow lost their linear predictive content, while Whale Activity ($\chi^2 = 20.03$, $p < 0.01$) gained significance. This pattern coincides with the Terra/Luna collapse in May 2022 and the FTX bankruptcy in November 2022, though our tests do not establish a causal link to these events. MVRV Ratio ($\chi^2 = 457.52$) and Open Interest ($\chi^2 = 190.43$) retained their dominant predictive roles, capturing valuation extremes and leverage dynamics during the downturn.

Our nonlinear analysis reveals the broadest set of statistically significant macro TE links in our sample. VIX (TE = 0.0228, $p < 0.01$), Gold (TE = 0.0282, $p < 0.01$), NASDAQ (TE = 0.0557, $p < 0.01$), Term Structure (TE = 0.0169, $p < 0.01$), and 10-year Treasury (TE = 0.0216, $p < 0.01$) all have a significant nonlinear causal relationship with BTC. This broadening of nonlinear macro linkages, which is more pronounced than in earlier periods, indicates that Bitcoin's response to macro conditions became increasingly state-dependent during the stress episode.

On-chain variables also show enhanced nonlinear dynamics. Exchange Reserves (TE = 0.0250, $p < 0.01$), Active Addresses (TE = 0.0260, $p < 0.01$), Funding Rate (TE = 0.0220, $p < 0.01$), MVRV Ratio (TE = 0.0719, $p < 0.01$), and Open Interest (TE = 0.0279, $p < 0.01$) transmit significant nonlinear information to BTC. Bidirectional nonlinear causality is evident for the MVRV Ratio (BTC→MVRV: TE = 0.0242, $p < 0.01$), suggesting feedback effects between valuations and returns that intensified during the deleveraging process.

5.5. Recovery (2024): Selective Re-engagement

The 2024 recovery phase, which coincides with the approval of spot Bitcoin ETFs in the United States, exhibits a selective reconfiguration of causality patterns. Linear tests (Table 7) show that macro-financial linkages narrowed: VIX ($\chi^2 = 25.49$) and NASDAQ ($\chi^2 = 22.41$) remain highly significant (both $p < 0.01$), while Gold, WTI Oil, and Term Structure lose their predictive power. The USD is significant at the 5% level ($\chi^2 = 6.19$) for longer cycles (34.3 days), suggesting a shift in the currency-exposure channel.

On-chain and derivatives metrics regained prominence as predictors. Whale Activity ($\chi^2 = 13.44$), Exchange Reserves ($\chi^2 = 19.40$), Exchange Netflow ($\chi^2 = 19.93$), Funding Rate ($\chi^2 = 23.72$), MVRV Ratio ($\chi^2 = 181.79$), and Open Interest ($\chi^2 = 148.51$) all display significant linear causality toward Bitcoin (all $p < 0.01$). The restoration of exchange-based predictors, combined with the continued strength of valuation and leverage metrics, reflects renewed speculative activity in this phase.

TE is significant for a wide set of on-chain and derivatives variables in the recovery period. There is significant nonlinear information transmission to BTC by Whale Activity (TE = 0.0350, $p < 0.01$), Exchange Reserves (TE = 0.0471, $p < 0.01$),

Exchange Netflow (TE = 0.0419, $p < 0.01$), Active Addresses (TE = 0.0323, $p < 0.01$), Funding Rate (TE = 0.0319, $p < 0.01$), MVRV Ratio (TE = 0.1840, $p < 0.01$), and Open Interest (TE = 0.0908, $p < 0.01$). It is in this period that the MVRV Ratio achieves its highest nonlinear effect across all periods, indicating that this valuation anchor operated through pronounced nonlinear channels during the recovery.

Among the macro-financial variables, NASDAQ (TE = 0.0505, $p < 0.01$) is the only significant nonlinear predictor, reinforcing Bitcoin's role as an equity-correlated, risk-sensitive asset. Bidirectional nonlinear causality is evident for Whale Activity (BTC→Whale: $p < 0.01$) and Funding Rate (BTC→Funding: $p < 0.01$). There is also a significant BTC→2Y Treasury channel (TE = 0.0245, $p < 0.01$), which is consistent with feedback from cryptocurrency returns to short-rate expectations proxies, though the reverse (2Y→BTC) is not significant.

5.6. Synthesis: Linear versus Nonlinear Causality Across Market Regimes

The joint analysis of frequency-domain Granger causality and TE analysis yields several observations about Bitcoin's evolving predictability structure. These patterns describe predictive relationships at a daily frequency; they do not establish structural transmission mechanisms. Linear and nonlinear approaches capture distinct dimensions of Bitcoin's information structure. Linear tests reflect the most transparent predictive structure in early regimes, while TE adds evidence of nonlinear dependence that is more apparent during boom–bust and stress regimes.

In the early Bitcoin period, nonlinear channels are limited (notably 10Y→BTC, BTC→NASDAQ, and BTC→MVRV). In the bull run period, TE analysis identifies several significant nonlinear crypto-native pathways (e.g., MVRV→BTC, Exchange Netflow→BTC, Hash Rate→BTC, and BTC→Funding Rate/Whale Activity). In contrast, the bear-market phase shows significant TE for multiple macro-financial variables, suggesting that non-linear coupling intensifies during periods of stress.

Bitcoin's predictability from traditional financial variables is stronger from 2019 onward, and especially pronounced in 2022 and 2023. Linear causality from the VIX, USD, and NASDAQ increased substantially between the early Bitcoin period and the bear market, though the pattern is not strictly monotonic (e.g., VIX→BTC is significant in the early Bitcoin phase, insignificant during the bull run, and highly significant thereafter). Nonlinear macro transmission is most pronounced in 2022 to 2023; outside that stress regime, it is limited, with the main exception of NASDAQ→BTC in 2024.

Despite the growing importance of macro-financial factors, on-chain variables retained substantial predictive power throughout all periods. The MVRV Ratio is consistently among the largest predictors: its linear statistic peaks in the institutional Adoption phase ($\chi^2 = 604.30$), while its TE effect peaks during the recovery (TE = 0.1840).

Exchange-based liquidity metrics and derivatives indicators (Funding Rate, Open Interest) complement the valuation signal, capturing the intensity of speculation and leverage dynamics. The persistence of these effects across regimes indicates that blockchain fundamentals provide predictive content that operates alongside, rather than displacing, macro-financial forces. The comparative analysis suggests that Bitcoin's predictability structure varies across market conditions. In lower-volatility environments, on-chain metrics exhibit greater incremental predictability via approximately linear channels, while macro-financial linkages remain subdued. In high-volatility or stress environments, nonlinear coupling with traditional markets intensifies, and feedback effects between Bitcoin and risk proxies become more prevalent. This pattern is consistent with regime-dependent price dynamics, though formal regime-switching tests would be needed to establish this.

The evidence does not support Bitcoin being a stable, safe haven. While the asset exhibited weaker macro-financial linkages during its early phase, the subsequent strengthening of predictability from equity markets, volatility indices, and interest rates positions Bitcoin as a liquidity-sensitive, high-beta asset rather than a store of value uncorrelated with traditional risk factors. The partial validity of the “digital gold” narrative is limited to the observation that on-chain fundamentals—analogous to supply-side factors in commodity markets—continue to have significant predictive power in valuations.

6. Conclusion

This paper has examined the evolving predictability structure of Bitcoin returns by jointly analyzing macro-financial variables and blockchain-native metrics within frequency-domain Granger causality and Tranger Entropy (TE) frameworks. Using daily data spanning from 2014 to 2024, we document how the information content of on-chain fundamentals and traditional financial indicators has shifted across five distinct market phases.

Three main findings emerge from our study. First, on-chain valuation and leverage metrics—particularly the MVRV ratio, funding rates, and open interest—have persistent predictive content for Bitcoin returns throughout all regimes examined. The MVRV ratio yields its highest linear test statistic during the 2019 to 2021 institutional adoption phase and its most significant nonlinear effect during the 2024 recovery. This persistence indicates that, as the market matured, blockchain-native indicators provide forecasting content that operates alongside, rather than being displaced by, macro-financial forces. Second, the predictability of Bitcoin based on traditional financial variables strengthened after 2019 and reached its peak during the monetary tightening episode from 2022 to 2023. Linear causality from the VIX, U.S. dollar index, and NASDAQ increased relative to earlier periods, though the pattern is not strictly

monotonic. The bear market phase produced the strongest macro-financial linkages in our sample, with equity indices and volatility measures exhibiting highly significant predictive relationships with Bitcoin at both short and medium-term frequencies.

Third, nonlinear methods reveal state-dependent predictability that is missed by linear approaches. TE from macro-financial variables to Bitcoin is largely insignificant before 2022, but broadens considerably during the bear-market stress episode. This asymmetry—limited nonlinear macro coupling in calmer periods, pronounced nonlinear transmission during stress—suggests that Bitcoin’s response to traditional market conditions intensifies in high-volatility environments. The 2024 recovery shows a partial normalization, with NASDAQ remaining the primary nonlinear macro predictor while on-chain metrics regain prominence.

These findings have implications for several audiences. For portfolio managers, the results indicate that Bitcoin’s diversification properties are regime dependent. The asset exhibited weaker macro-financial linkages in its early years, but after 2019, the evidence points to its substantial comovement with equity markets and sensitivity to monetary conditions, particularly during periods of stress. Allocators should not assume stable correlation structures when constructing portfolios that include Bitcoin. For risk managers, the nonlinear results highlight the fact that standard linear models may underestimate tail dependence during market dislocations. The intensification of TE from volatility indices and equity markets in 2022 and 2023 suggests that Bitcoin’s exposure to traditional risk factors increases precisely when diversification would be most beneficial. Stress-testing frameworks should account for this state-dependent behavior.

For policymakers and regulators monitoring financial stability, two observations merit attention. The persistent predictive content of on-chain metrics—exchange flows, funding rates, and leverage proxies—indicates that blockchain transparency allows the use of monitoring tools unavailable for traditional assets. At the same time, the strengthening of macro-financial linkages implies that cryptocurrency markets are no longer isolated from broader financial conditions. Spillover risks may flow in both directions, particularly as institutional participation deepens through ETFs and regulated derivatives.

Several limitations warrant acknowledgment. Our analysis documents predictive relationships in the Granger sense; we do not claim to identify structural transmission mechanisms. The five-phase periodization, while grounded in the structural break analysis and major market events, involves making judgment calls that could be specified differently. TE results are sensitive to embedding parameters and may capture dynamics different from those captured by alternative nonlinear approaches. Finally, the 2024 recovery phase contains fewer observations than earlier periods, and patterns observed in this regime may not persist as the market evolves.

Table 3: Frequency-Domain Granger Causality (Daily) — Early Bitcoin (2014-10-01 to 2016-12-31)

Variable	Obs	To BTC			From BTC			
		N	Chi ²	Sig	Period (d)	Chi ²	Sig	Period (d)
<i>Market variables</i>								
VIX	554	9.04	**		2.0	0.69		4.1
US Dollar	554	1.35			5.5	5.21	*	2.1
Gold	554	1.93			3.2	3.36		16.9
NASDAQ	554	11.20	***		2.0	2.45		628.3
WTI Oil	554	10.99	***		3.2	11.84	***	2.8
Term Structure	554	7.17	**		2.1	4.89	*	2.8
10Y Treasury	554	5.66	*		2.1	2.01		8.4
<i>On-chain variables</i>								
Whale Activity	821	6.70	**		2.2	3.95		2.5
Exchange Reserves	821	16.69	***		3.0	10.65	***	24.0
Exchange Netflow	821	13.07	***		3.0	5.54	*	2.5
Active Addresses	821	2.55			3.7	8.06	**	37.5
Hash Rate	821	3.13			6.6	4.70	*	31.6
Transaction Fees	821	4.91	*		2.1	11.29	***	628.3
Funding Rate	230	8.84	**		628.3	22.87	***	9.8
MVRV Ratio	821	14.57	***		34.3	4.53		3.4

Sample. Market: 2014-10-02 to 2016-12-30 (business days = 554). On-chain: 2014-10-02 to 2016-12-31 (observations = 821; Funding Rate = 230).

Notes. Breitung-Candelon frequency-domain Granger causality; period in days (period = $2\pi/\omega$). Significance: * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 4: Frequency-Domain Granger Causality (Daily) — Bull Run Era (2017-01-01 to 2018-12-31)

Variable	Obs	To BTC			From BTC			
		N	Chi ²	Sig	Period (d)	Chi ²	Sig	Period (d)
<i>Market variables</i>								
VIX	494	2.08			7.7	5.69	*	2.3
US Dollar	494	1.97			628.3	5.97	*	4.5
Gold	492	5.17	*		3.5	4.41		2.0
NASDAQ	494	4.68	*		15.0	5.18	*	2.2
WTI Oil	494	2.25			3.9	1.03		5.9
Term Structure	494	5.16	*		16.2	6.20	**	2.8
10Y Treasury	494	1.52			2.2	6.32	**	2.8
<i>On-chain variables</i>								
Whale Activity	729	1.57			14.4	10.78	***	21.4
Exchange Reserves	729	16.83	***		5.5	5.55	*	3.9
Exchange Netflow	729	13.03	***		7.2	4.44		3.4
Active Addresses	729	19.02	***		628.3	27.13	***	29.3
Hash Rate	729	7.36	**		4.7	8.09	**	34.3
Transaction Fees	729	4.34			4.5	7.06	**	628.3
Funding Rate	729	9.07	**		4.7	42.87	***	18.5
MVRV Ratio	729	72.05	***		3.4	4.59		4.5

Sample. Market: 2017-01-04 to 2018-12-31 (business days = 494). On-chain: 2017-01-02 to 2018-12-31 (observations = 729).

Notes. Breitung-Candelon frequency-domain Granger causality; period in days (period = $2\pi/\omega$). Significance: * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 5: Frequency-Domain Granger Causality (Daily) — Institutional Adoption (2019-01-01 to 2021-12-31)

Variable	Obs	To BTC			From BTC			
		N	Chi ²	Sig	Period (d)	Chi ²	Sig	Period (d)
Market variables								
VIX	743	32.26	***		15.0	5.96	*	3.7
US Dollar	743	34.35	***		2.2	6.54	**	2.3
Gold	743	14.87	***		7.7	4.42		21.4
NASDAQ	743	27.93	***		2.3	4.76	*	52.3
WTI Oil	743	7.93	**		2.2	2.11		2.7
Term Structure	743	5.67	*		2.0	2.07		14.4
10Y Treasury	743	7.71	**		2.1	1.68		20.3
On-chain variables								
Whale Activity	1095	12.59	***		19.3	3.26		8.7
Exchange Reserves	1095	5.95	*		628.3	6.30	**	3.0
Exchange Netflow	1095	4.58			3.2	7.42	**	8.4
Active Addresses	1095	8.20	**		21.4	23.27	***	22.6
Hash Rate	1095	5.95	*		6.2	4.99	*	31.6
Transaction Fees	1095	5.42	*		6.0	11.04	***	4.3
Funding Rate	1095	52.06	***		31.6	3.37		3.7
MVRV Ratio	1095	604.30	***		21.4	7.05	**	2.5
Open Interest	1007	145.98	***		21.4	12.73	***	2.5

Sample. Market: 2019-01-03 to 2021-12-30 (business days = 743). On-chain: 2019-01-02 to 2021-12-31 (observations = 1095; Open Interest = 1007).

Notes. Breitung–Candelon frequency-domain Granger causality; period in days (period = $2\pi/\omega$). Significance: * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 6: Frequency-Domain Granger Causality (Daily) — Bear Market (2022-01-01 to 2023-12-31)

Variable	Obs	To BTC			From BTC			
		N	Chi ²	Sig	Period (d)	Chi ²	Sig	Period (d)
Market variables								
VIX	496	54.00	***		2.8	10.52	***	6.1
US Dollar	496	27.98	***		4.3	2.77		25.5
Gold	496	8.72	**		3.6	1.50		6.5
NASDAQ	496	98.94	***		2.3	6.06	**	2.0
WTI Oil	496	4.04			2.8	2.21		2.1
Term Structure	496	6.43	**		628.3	3.25		2.1
10Y Treasury	496	16.00	***		27.3	0.52		22.6
On-chain variables								
Whale Activity	729	20.03	***		2.5	8.43	**	4.8
Exchange Reserves	729	2.96			4.8	3.12		628.3
Exchange Netflow	729	2.06			4.8	0.72		3.3
Active Addresses	729	1.20			11.9	2.37		2.0
Hash Rate	729	2.48			7.3	7.55	**	2.0
Transaction Fees	729	2.26			5.9	14.94	***	2.4
Funding Rate	729	3.42			2.5	5.43	*	2.1
MVRV Ratio	729	457.52	***		21.4	3.52		2.5
Open Interest	729	190.43	***		20.3	10.93	***	21.4

Sample. Market: 2022-01-04 to 2023-12-29 (business days = 496). On-chain: 2022-01-02 to 2023-12-31 (observations = 729).

Notes. Breitung–Candelon frequency-domain Granger causality; period in days (period = $2\pi/\omega$). Significance: * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 7: Frequency-Domain Granger Causality (Daily) — Recovery (2024-01-01 to 2024-12-31)

Variable	Obs	To BTC			From BTC			
		N	Chi ²	Sig	Period (d)	Chi ²	Sig	Period (d)
Market variables								
VIX	249	25.49	***		2.3	5.42	*	3.6
US Dollar	249	6.19	**		34.3	3.48		27.3
Gold	249	3.31			2.8	2.27		2.7
NASDAQ	249	22.41	***		2.3	6.51	**	2.0
WTI Oil	249	1.27			5.1	2.53		2.2
Term Structure	249	0.18			4.0	3.53		4.5
10Y Treasury	249	2.61			6.3	7.75	**	2.1
On-chain variables								
Whale Activity	365	13.44	***		20.3	1.36		2.5
Exchange Reserves	365	19.40	***		3.4	0.84		5.4
Exchange Netflow	365	19.93	***		3.5	1.34		6.9
Active Addresses	365	3.15			3.8	1.92		21.4
Hash Rate	365	3.25			2.1	3.70		628.3
Transaction Fees	365	1.93			3.8	2.53		2.0
Funding Rate	365	23.72	***		2.8	6.95	**	2.1
MVRV Ratio	365	181.79	***		20.3	6.26	**	4.0
Open Interest	365	148.51	***		19.3	7.43	**	7.4

Sample. Market: 2024-01-03 to 2024-12-31 (business days = 249). On-chain: 2024-01-02 to 2024-12-31 (observations = 365).

Notes. Breitung-Candelon frequency-domain Granger causality; period in days (period = $2\pi/\omega$). Significance: * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 8: Transfer Entropy (TE) between Variables and Bitcoin by Period. Stars denote significance: $*p<.10$, $**p<.05$, $***p<.01$. Periods: Early Bitcoin (2014–2016), Bull Run Era (2017–2018), Institutional Adoption (2019–2021), Bear Market (2022–2023), Recovery (2024).

Variable	Early Bitcoin		Bull Run Era		Institutional Adoption		Bear Market		Recovery	
	var→BTC	BTC→var	var→BTC	BTC→var	var→BTC	BTC→var	var→BTC	BTC→var	var→BTC	BTC→var
<i>On-chain variables</i>										
Whale Activity	0.0082	— 0.0086	0.0129	— 0.0285***	0.0072	— 0.0082	0.0119	— 0.0137	0.0350***	— 0.0295***
Exchange Reserves	0.0135	— 0.0089	0.0078	— 0.0168	0.0099	— 0.0106	0.0250***	— 0.0130	0.0471***	— 0.0145
Exchange Netflow	0.0050	— 0.0111	0.0212***	— 0.0048	0.0066	— 0.0121	0.0135	— 0.0153	0.0419***	— 0.0229
Active Addresses	0.0093	— 0.0039	0.0060	— 0.0087	0.0043	— 0.0068	0.0260***	— 0.0108	0.0323***	— 0.0177
Hash Rate	0.0105	— 0.0052	0.0220***	— 0.0114	0.0178***	— 0.0023	0.0028	— 0.0133	0.0215	— 0.0208
Transaction Fees	0.0059	— 0.0057	0.0084	— 0.0167	0.0139	— 0.0054	0.0126	— 0.0054	0.0217	— 0.0149
Funding Rate	0.0170	— 0.0427	0.0126	— 0.0192***	0.0174***	— 0.0115	0.0220***	— 0.0088	0.0319***	— 0.0227***
MVRV Ratio	0.0139	— 0.0141***	0.0755***	— 0.0194	0.0909***	— 0.0090	0.0719***	— 0.0242***	0.1840***	— 0.0148
Open Interest	— — —		— — —		0.0549***	— 0.0091	0.0279***	— 0.0096	0.0908***	— 0.0178
<i>Market variables</i>										
VIX	0.0075	— 0.0153	0.0077	— 0.0115	0.0114	— 0.0112	0.0228***	— 0.0210	0.0201	— 0.0116
US Dollar (DXY)	0.0082	— 0.0069	0.0161	— 0.0075	0.0070	— 0.0139	0.0108	— 0.0097	0.0270	— 0.0111
Gold	0.0079	— 0.0023	0.0161	— 0.0064	0.0097	— 0.0059	0.0282***	— 0.0092	0.0173	— 0.0084
NASDAQ	0.0059	— 0.0193***	0.0139	— 0.0047	0.0114	— 0.0149	0.0557***	— 0.0156	0.0505***	— 0.0119
WTI Oil	0.0171	— 0.0065	0.0130	— 0.0074	0.0066	— 0.0077	0.0232	— 0.0101	0.0319	— 0.0215
Term Structure	0.0154	— 0.0078	0.0108	— 0.0107	0.0118	— 0.0109	0.0169***	— 0.0047	0.0185	— 0.0277
10Y Treasury	0.0187***	— 0.0138	0.0186	— 0.0183	0.0219***	— 0.0054	0.0216***	— 0.0088	0.0316	— 0.0214
2Y Treasury	0.0132	— 0.0122	0.0155	— 0.0110	0.0117	— 0.0103	0.0099	— 0.0238***	0.0323	— 0.0245***

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